

UGED1111 Logic

CATEGORICAL SYLLOGISM I

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Validity

Valid argument(對確論證):

◆ It is impossible that all its premises are true and the conclusion is false.

Conversely,

Invalid argument:

◆ It is possible that all the premises are true and the conclusion is false.

Question 1



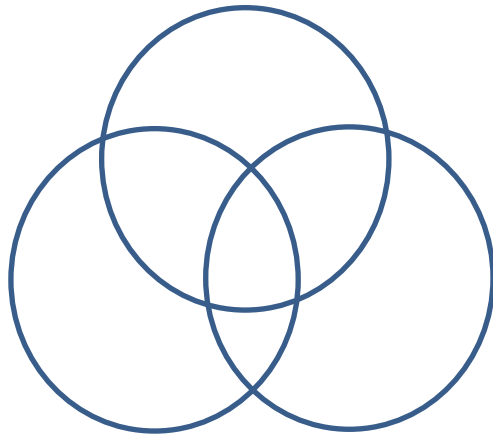
Someone is sick.

Someone is unhappy.

Someone is sick and unhappy

Any tools for checking validity?

➤ Categorical Syllogism



Agenda



- **Categorical statement (/proposition)**
- **Quality and Quantity**
- **Boolean interpretation Vs Aristotelian interpretation**
- **Venn Diagram**

Categorical Statement

Categorical statement

- ◆ A statement that relates two classes, or categories, is called a categorical proposition.
- ◆ The classes in question are denoted respectively by the subject term and the predicate term.

Examples



1. 靚仔無本心。
2. 俄羅斯人是戰鬥民族。
3. 所有稱為三文魚的魚都可以生食。
4. All cats catch mice.
5. Some coke is designed for people on a diet.

Standard Categorical Statement

Components:

Quantifier: *“All” ; “Some” ; “No”*

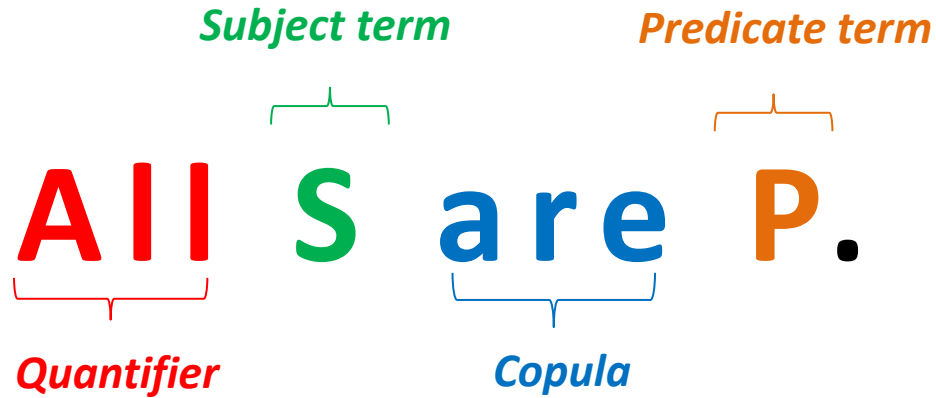
Subject term

Predicate term

Copula: *the words link Subject term and*

Predicate term, i.e. “are”, “are not”

Standard Categorical Statement



Types of Categorical Statement

- (1) those that assert that the whole subject class is included in the predicate class,
- (2) those that assert that part of the subject class is included in the predicate class,
- (3) those that assert that the whole subject class is excluded from the predicate class, and
- (4) those that assert that part of the subject class is excluded from the predicate class.

Standard Forms of Categorical statement

Standard form:

- (1) All S are P.**
- (2) Some S are P.**
- (3) No S are P.**
- (4) Some S are not P.**

Not standard:

Example

- 1. All S are not P.**
- 2. Some Not-S are P.**

Quality and Quantity

Quality

- affirmative quality:

"All S are P" and "Some S are P"

- negative quality:

"No S are P" and "Some S are not P" .

Quality and Quantity

Quantity

◆ **Universal : "All S are P" and "No S are P"**

e.g. All cats are mammal.

No whales are fishes.

◆ **Particular: "Some S are P" and "Some S are not P"**

E.g. Some people are bad persons.

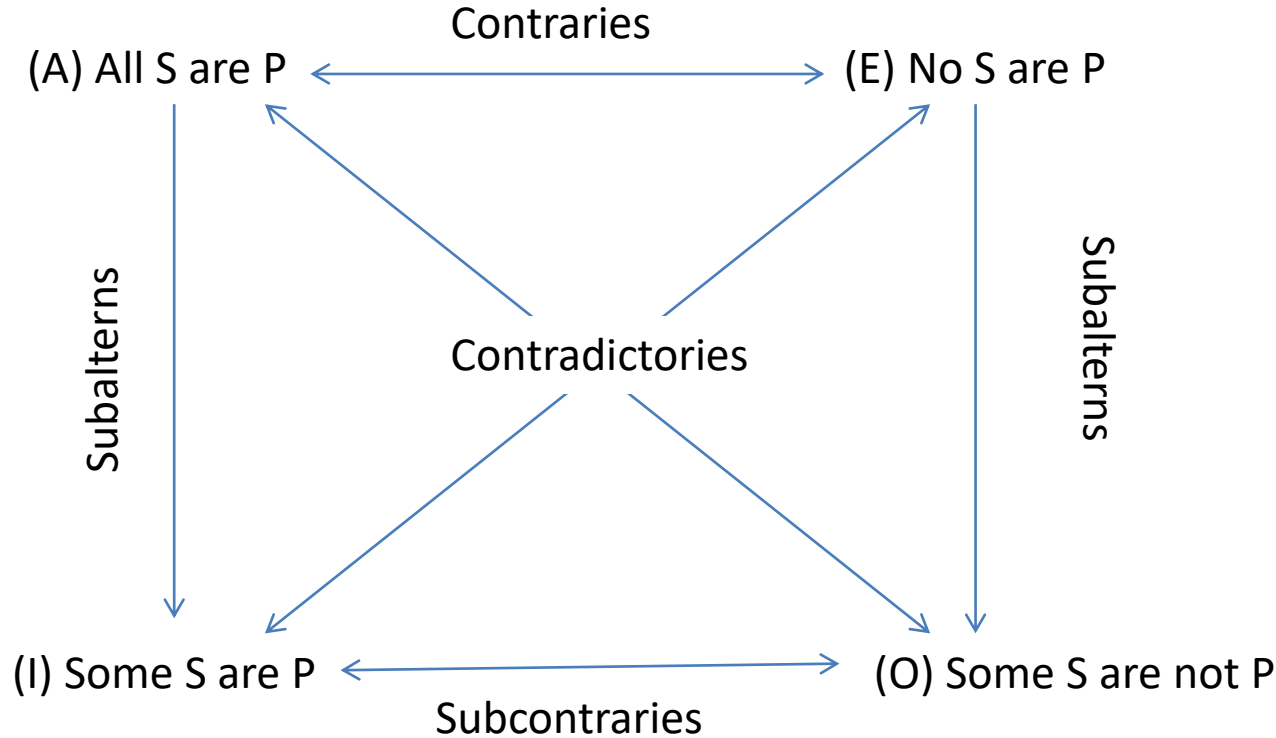
Some insects are not lovely animals.

A,E,I,O

- The universal affirmative is called an **A Proposition**
- the universal negative is called an **E proposition**
- the particular affirmative is called an **I proposition**
- the particular negative is called an **O proposition**.

Proposition	Letter Name	Quantity	Quality
All S are P.	A	Universal	Affirmative
No S are P.	E	Universal	Negative
Some S are P.	I	Particular	Affirmative
Some S are not P.	O	Particular	Negative

The traditional square of opposition



The traditional square of opposition

Relation	Definition
Contradictory	opposite truth value
Contrary	at least one is false (not both true)
Subcontrary	at least one is true (not both false)
Subalternation	truth flows downward; falsity flows upward

The traditional square of opposition

- “All S are P” $\rightarrow$ “Some S are P”
- “No S are P” $\rightarrow$ “Some S are not P”
- “Some S are P” and “Some S are not P” can not both be false.
 - There is at least one S, so this S is either P or not P.

Example

All cats have a tail.



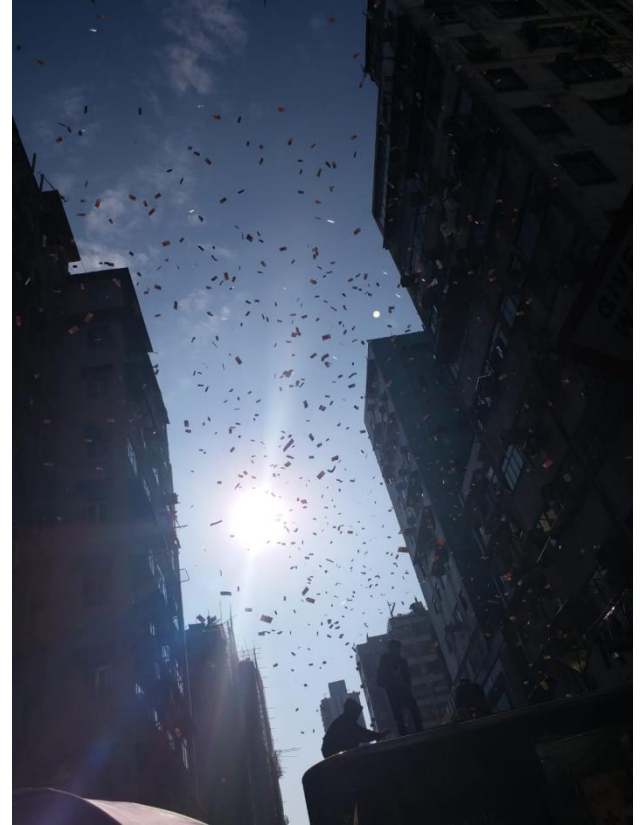
Some cats have a tail.



Scenario 1

Imagine someone told you that
“Come! All banknotes on the floor of
NAH11 are welcome to be collected”

However, you found no banknotes
when you come to class.



Scenario 2



Imagine a lecturer told you that
“All students who fail the course are who
need to retake the course.”

Do the lecturer presuppose there is at
least one student need to retake?

Existential import



For **natural language**:

- We have existential import for some “all”-statement
- Depend on context!

How about formal language?

Aristotelian Standpoint

Aristotelian: *“All” implies “some”*

“All students who fail the course are who need to retake the course.”

Implies

“Some students who fail the course are who need to retake the course.”

Interpretations

All dragons have wings.



Some dragons have wings.

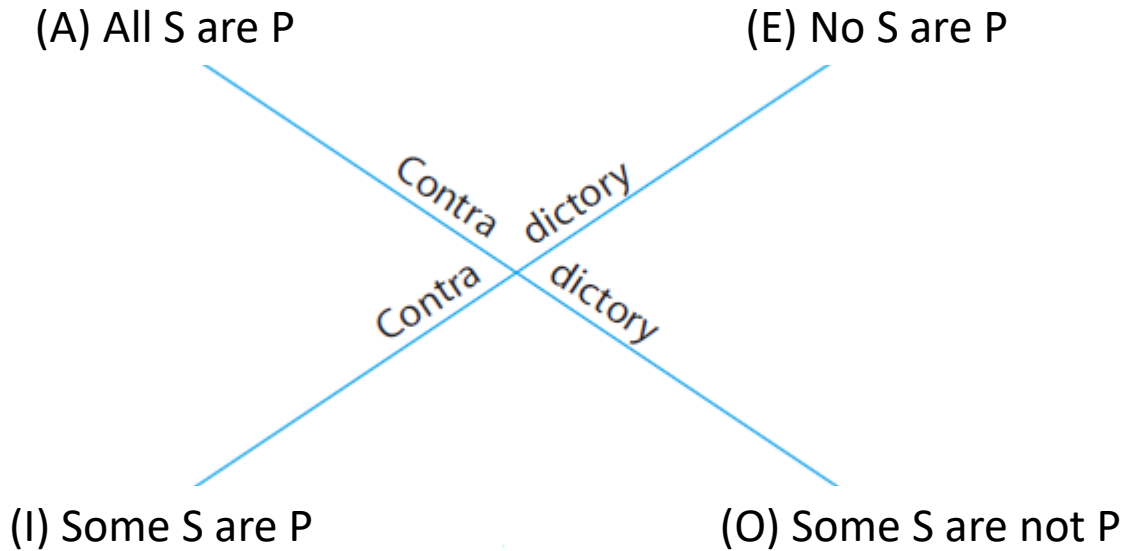


所有獨角獸有角。



有些獨角獸是有角。

Modern Square



Boolean Standpoint

Boolean:

“All students who fail the course are who need to retake the course.”

doesn't imply

“Some students who fail the course are who need to retake the course.”

“If there is such a thing as a student who fails the course , it is a person who need to retake the course.”

Existential Fallacy

1. All cats are animals.
2. Some cats are animals.

Boolean: Invalid, existential fallacy
Aristotelian: Valid

**No existential import
for empty terms for
Aristotelian standpoint!**

1. All unicorns are animals.
2. Some unicorns are animals.

Boolean: Invalid, existential fallacy
Aristotelian: Invalid, existential fallacy

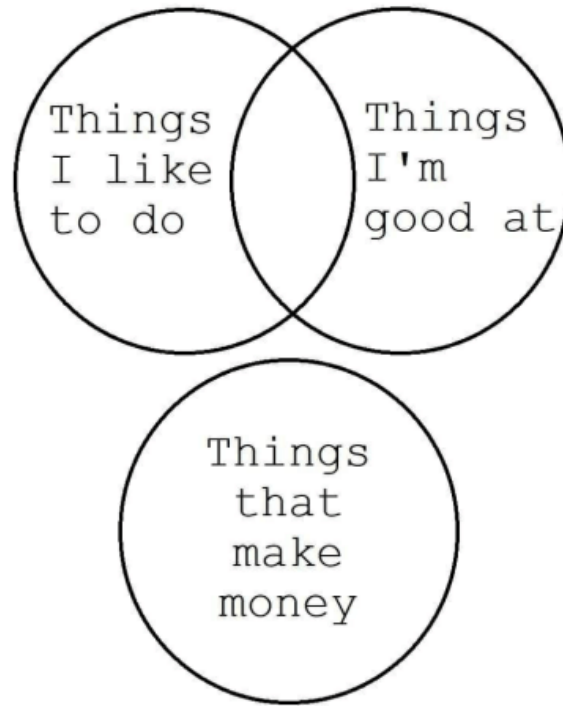
Aristotelian Vs Boolean

For non-empty terms

	Existential import in "All" / "No" statement	Existential import in "Some" statement
Aristotelian	Yes	Yes
Boolean	No	Yes

How to represent a categorical statement?

Venn Diagram of my life:

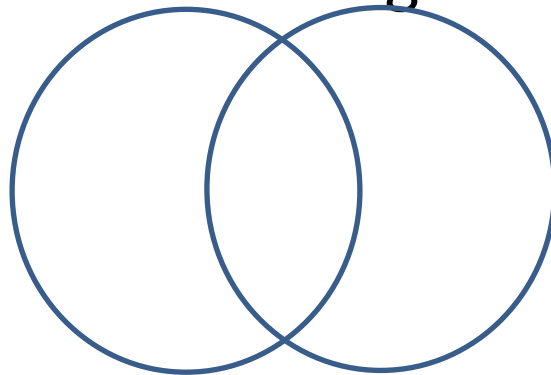


Venn Diagram

Represent the relationship of two terms within a categorical statement.

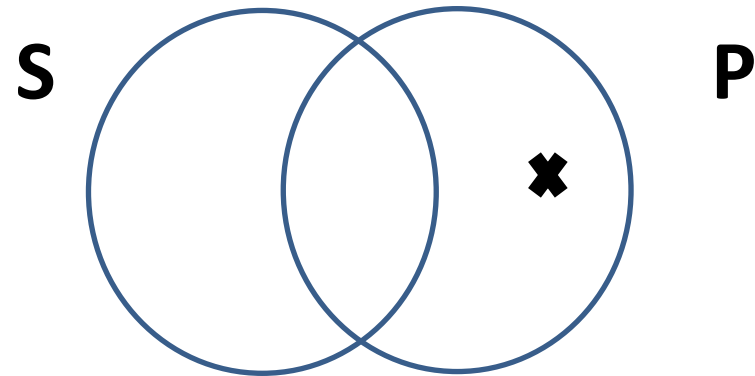
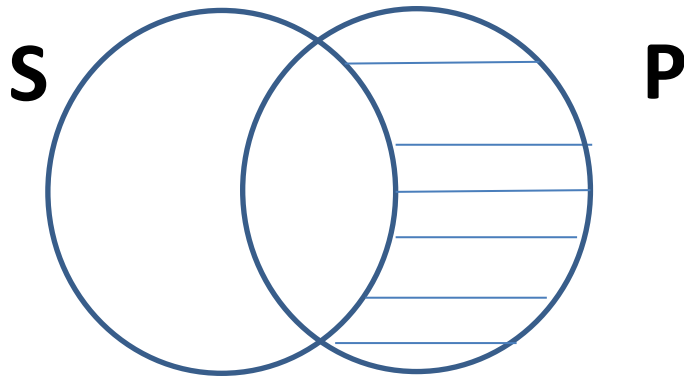
How to represent the four kinds of categorical proposition on a Venn Diagram

- Start with two intersecting circle as follows



Venn Diagram

- Two kinds of marks are used: shading an area and placing an "X" in an area.
 - Shading an area means that the shaded area is empty,
 - placing an "X" in an area means that at least one thing exists in that area.

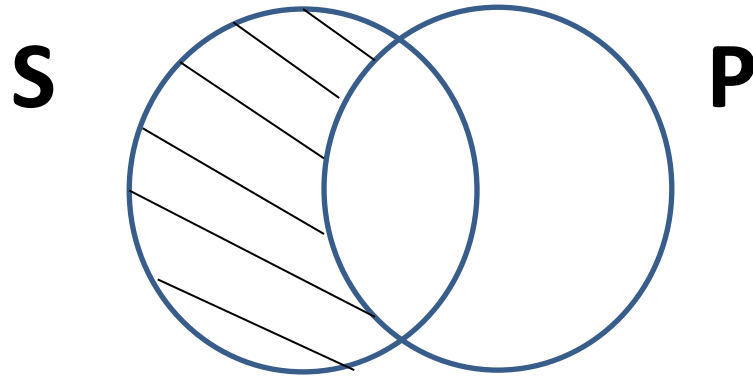


Drawing Venn Diagram

- Shading an area means that the shaded area is empty
 - For **universal** statements (*“all” or “no”*)
- Placing an "X" in an area means that at least one thing exists in that area
 - For **particular** statements (*“some”*)

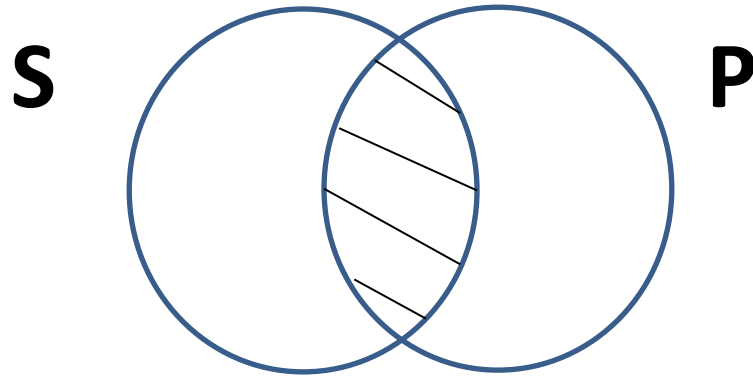
Venn Diagram

All S are P.



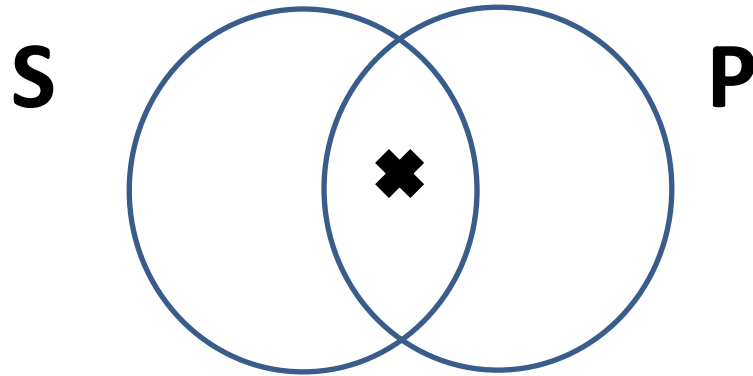
Venn Diagram

No S are P.



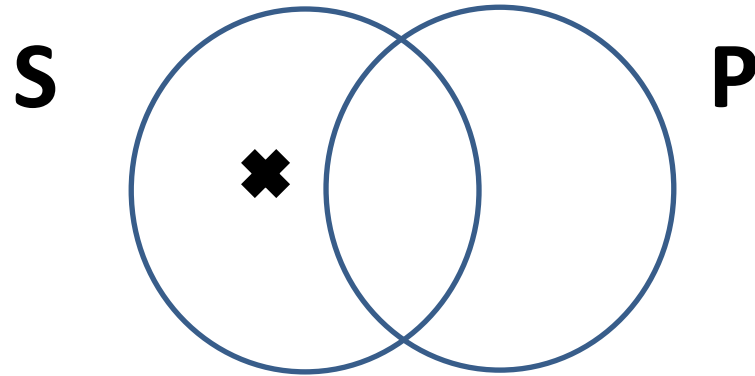
Venn Diagram

Some S are P.

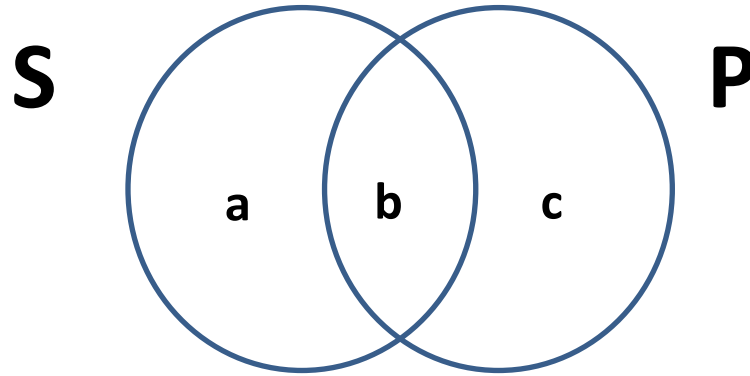


Venn Diagram

Some S are not P.



Venn Diagram



A: Shade a

B: Shade b

C: Shade c

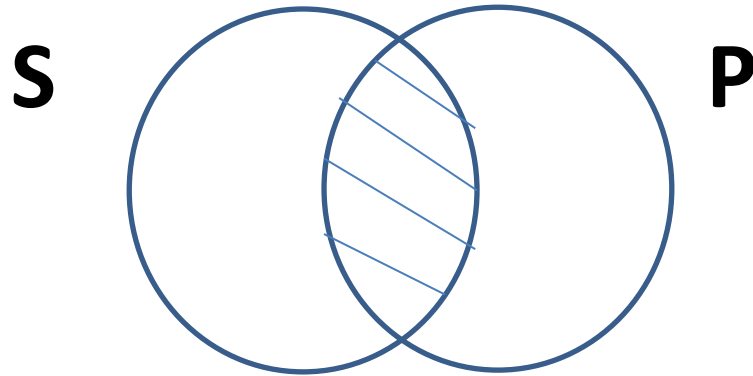
D: Mark X on a

E: Mark X on b

F: Mark X on c

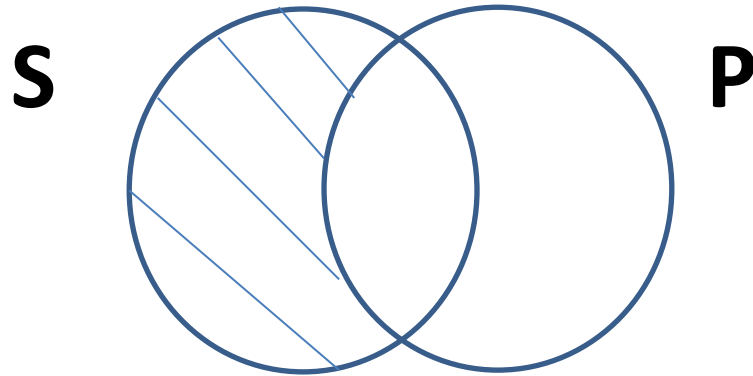
Discussion 1

All S are not P.
= No S are P



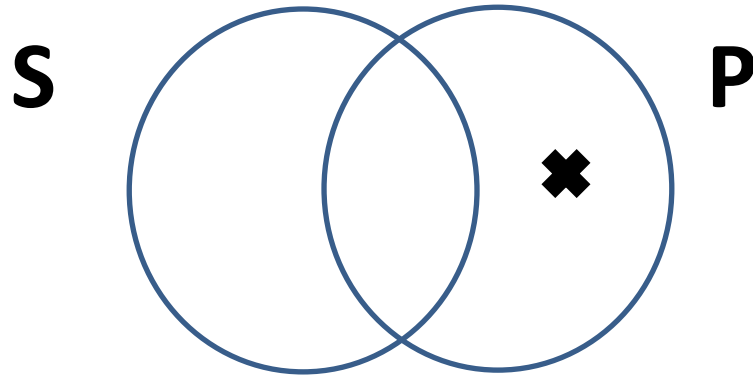
Discussion 2

No S are not P.
= All S are P.



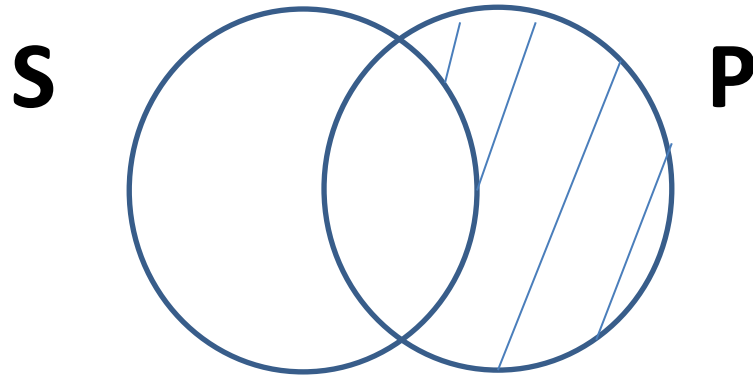
Discussion 3

Some non-S are P.
= Some P are not S.



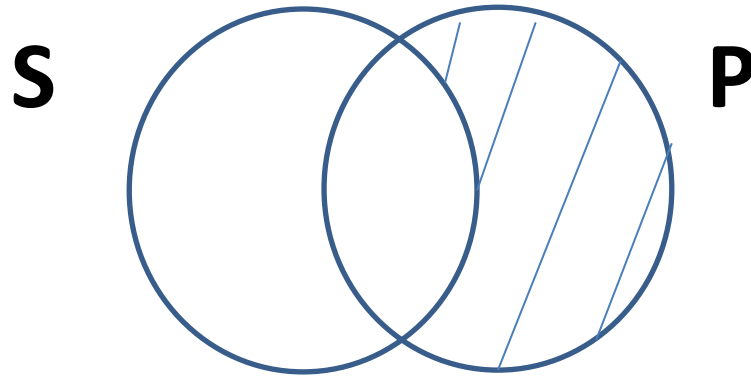
Discussion 4

All non-S are not P.
= All P are S.



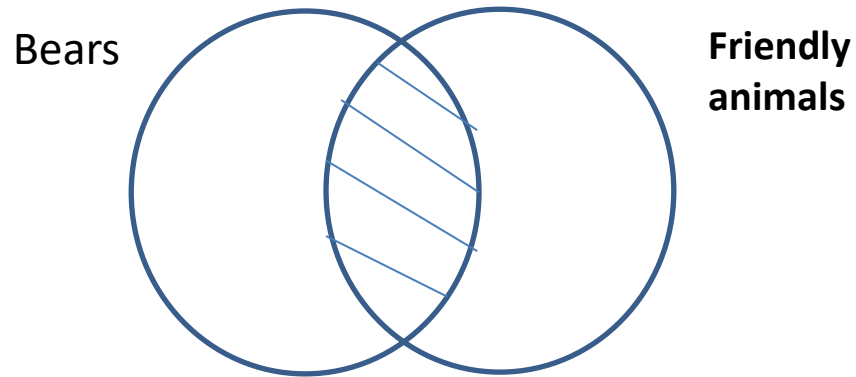
Discussion 5

It is not the case that Some P are not S .
= All P are S.



Discussion 6

No bears are friendly animals.



Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.